

Misleading with Charts — How Bad Visualisation Can Mislead in Decision Making Transcript

Hello everyone, welcome to BITS PLANET DIGITALS. Today we will continue our learning journey for the force data visualisation storytelling and for the module building dashboards. Today we will be particularly focussing on understanding what are the different types of charts that can at times be misleading and end up as bad visualisations, though the intention was not such.

And these bad visualisations or charts with misleading characters can impede the decision making and deriving the entire purpose of data visualisation. So recall the main purpose of data visualisation exercise is to convey information in the most efficient format to your audience and enable them to absorb the insights in the data and take actions that you want them to take. So now let us look at the chart that is there on the slide.

We humans are very respond to visual perceptions very fastly. So there is some amount of natural logic that works when we perceive a graphic. For example, when you see the graphic that is there on the slide, you would immediately see that there is a fall in the number of incidents or the graph is showing a decreasing trend because there is some amount of downward trend after a particular event.

But in this case, the problem was the graph portrays the number of murders committed using firearms after a particular law is enacted in Florida. We are not interested in the particular law per se, but what we are trying to look at is the way the graphic is depicted. If you see at the initial look, you would notice the graph is showing a falling trend.

Somehow the law is enacted in 2005 and subsequently the number of murders are moving and there is a fall in number of murders. But the problem here is the axis is inverted in the sense that it goes from 0 to 900 in the opposite direction which is counterintuitive to our natural ordering. Generally, we are used to see graphs where the lowest number starts at the below and it goes in an ascending order.

So, when you look at this graph, your attributes like the colour and the position of the axis drives you to a conclusion that there has been a decrease in the fatalities after a particular act is enacted in Florida. But in fact, the conclusion is quite the opposite. The reasoning that the analyst who has presented this graph has given is that murders indicates a negative sentiment and therefore the colour red and the decrease in murders should be seen as a better point.

But if you were to see, there is actually increase in the number of murders from 500 to 800, almost 850 after the enactment of law. This is not deceived when we see the graph directly and we would naturally think that the number of murders have

decreased after the enactment of law. So, one of the cautions that we can take from this graph is that one should not try to go against the natural ordering that we humans are used to.

And in case you want to convey this message more emphatically, you can use some colour though the axis could have been normal. You can use red colour to show that these are murders by use of firearm. So, we can use colours to bring in the effect but should not disturb the natural logical ordering.

Though the chart was informative, at the first glance you would end up drawing the exact opposite conclusion that it wanted to convey. So, we are talking about axis manipulation. Many a times, analysts out of zeal to present or overemphasise on a particular topic, particular data point they would try to truncate y-axis and make comparisons either showing the difference to be much larger than they are or much lower than they are.

They also remove the baseline to exaggerate the difference between two points and we will see couple of examples and understand this. So, what is the lesson from this? Always check the axis scale while you are looking at any visualisation and the lesson for the analyst is you should not purposefully change the axis and in some sense manipulate the presentation of the data. So, look at the examples that you have on the chart.

The first chart shows the change in the tax rate. Currently, the tax rate is 35% and it is going up to 39.6% and for any visual person, once you look at directly it shows there is a very steep increase between the tax rates. But if you notice, only the x-axis or the y-axis has been presented from 34% to 40% talking about a range of 6% and therefore you see this big differential.

But in fact, when the actual scale is presented which is given in the centre, you could see the relative increase is much smaller and it has to be compared with the base which is 0. So, while truncating the y-axis or removing the 0 baseline, they exaggerate the differences. So, this is another way to unnecessarily zoom out the y-axis like 34% to 42%. Only focus on a particular area and show the difference by altering the scale.

So, again all these graphs are presenting the same quantity when the actual growth is from 35% to 39.6%. In the first and the last graph, the differences are exaggerated or blown out of proportion where the actual presentation is given in the middle. So, again while checking any visual you have to pay attention to the axis as a reader and as an analyst you should take care not to manipulate the axis. So, we will now see another issue called visualisation non-septotorque which we use decoration or bring out additional elements into the graph and deep down you are dumping down the message that the visualisation wants to convey.

For example, look at the graphs that are on the left side of the slide. So, it talks about a survey and how many times recreational drugs or alcohol has been used by a college. We are not interested in the details per se, but anybody looking at the graph feels that there has been sudden increase between the two time periods in which the survey has been done.

But one can clearly also understand the scale is not comparative to the levels that are presented on the graph. Again, this is the case of axis manipulation where they are trying to fix the axis across all the graphs that are there in the visual and it clearly shows 0.7 and 29.4. The order of difference is much more higher but they are presented as if they are comparable. So, we do not know the actual scales that are used for presenting these graphs and this can at times lead to misleading of the graph at sometimes trying to reduce the information, deemphasizing the information that we want to convey or overemphasizing the information that we want to get across.

The same thing is also done on the graph that is on the right side where the increase in petrol price has been shown using the bar graphs and they suddenly show the petrol price has fallen by 28%. That is the main message the graph is trying to convey using the pre-attentive attributes and the visual features like there is an arrow mark showing increase and arrow mark showing decrease and the numbers that are highlighted clearly to bring the capital. And the fine print, the main thing is actually the price has increased.

It is 72.83 compared to 56.71. Yes, the main point was the rate of increase between the two periods is much lesser compared to the previous periods. Nobody is denying that but presenting the graph with reduced bar length so remember when we have studied in earlier lessons the way that humans perceive any visual is through to look at the physical assembly of the geographical elements, decode assembly. So, they are trying to understand the length of the bar being representative of the petrol prices.

In this case, the fall in the petrol prices is not there. Actually petrol prices have increased but what has fallen is the rate of increase of petrol price. The rate at which petrol price has changed.

From 21 it has gone to 30 rupees, 31 rupees 42% increase. From 30 it has gone to 56. Almost 83% increase or 84% increase.

But now it has gone to 72 rupees which is only 30% increase. So, the graph should have been a linear graph where the bar should have been somewhere here whereas the increase they could have emphasised on the rate of increase. So, in some sense the graph is misleading.

The next issue that we face while using graphs and how people can mislead the audience is using the scaling issues or dual access. For example, when you are

comparing quantities using line graphs care has to be taken as the scale of the units that are presented on the y-axis so that comparisons can be done clearly. Because of the scale of cancer screenings that are done is very high compared to the abortions which are presented on the same scale.

It looks like there is a linear trend in abortions when in fact the rate of increase the nature of the slope of the curve is lost because of the scaling. So, we have to use appropriate scaling to ensure we are able to present both the trend and the quantity effect on the visual. And in this case you can use the dual access chart, use dual access clearly label them so that audience are not confused.

So, again dual access can also be misused if you are not giving the scales clearly it can lead to long comparisons. So, dual access is a very powerful tool but at the same time unless you provide clarity on the scales that are on both the axis you might end up misleading the audience. And now we come to a very important concept which an analyst should always bear in his mind that while we explore lot of data patterns and one of the major steps of data visualisation exercise.

If you recall the seven steps of data visualisation is understanding the underlying patterns in the data in which you would plot lot of scatter diagrams, line diagrams to understand the relationship between two or three variables and at times it shows it might turn out that some of these variables are correlated for whatever reasons that we have. For example, the number of murders are in some sense following the same trend as the ice cream sold or to any natural I mean to any common sense there should not be any reason why one is leading to other or do murders lead to more ice cream sales or ice cream sales leads to more murders but they are just correlated. They have some linear trend together but we cannot per se say one is causing the other.

So, it is always important to understand and remember back of our minds that correlation does not mean causation. We need to always be very careful of overstretching our conclusions by saying the correlation that we found might be an indicative of the causation. So, we have to be very very careful why we are enthusiastically exploring relationship between the variables in our dataset using visualisations and portraying them.

We should start we should stop short of calling any causal relationship between them unless we know there is some theoretical explanation or some business logic behind that. So, there is one more graph at the below which shows that Nicolas Cage movies versus drowning in swimming pools follow a similar trend which means they are more correlated but can we ask can we say that they are statistically leading I mean there is a causal link between them which is statistically significant. So, there might be some confounding variable which is influencing these two variables together which we have not accounted for in our analysis.

Once we get the confounding variable the relationship might change sometimes it is not possible to know what the confounding variable is also so what we try to do is like present the correlation and stop short of calling the causality between the variables. We will now look at something called Simpson's Paradox which talks about how you group the data like within a visual or while presenting a data analysis there might be subsets or what we call finer divisions in data. Let me give you an example.

When you are talking about business of regions which is a common use case in the business parallel. You are a business entity you have different regions and each region has different products. Products have different stores each store sells different products and you group them.

So, the way you group the data might convey a different message if you follow a random order or if you change the grouping order you might get a different message. Let me give you an example for example there are two people who are doing tasks on Saturday and Sunday so the person A completes 7 out of 8 tasks on Saturday while person B completes 2 out of 2 tasks on Saturday so if you see person A has done 87.5% of the job person 2 has done 100% of the job. Come Sunday the person 2 does only 1 out of 2 jobs assigned to him which is 50% and person B has done only 5 out of 8 jobs that is 62.5. Clearly on both days person A or store A or region A has done better than region B has done better than region A. But when you combine overall so it is 80% success rate for person A or efficiency completion for person A versus person B which turns out to be 70%.

So if I start looking at overall number and without getting into the finer details I might draw different conclusions if I look at the finer details I might draw different conclusions. So grouped data can be deceptive so one needs to be careful how the group composition is arrived at and what are the finer components within that group hierarchy and if needed drill down and this is where the interactivity of dashboards becomes important. Though as an analyst you would want to present the overall picture you should enable the user to have the interactivity and the ability to drill down and see the underlying components.

So this also fits into the building a narrative which is both author driven and the user driven. So again why do we enable drill down? Again the order of filtering matters following the same logic that we have discussed a while ago. The way we group also enables us to drill down in a particular fashion.

If you want to drill down if you allow a drill down sequence which is preset you are actually driving down certain form of conclusions. Let us understand this by taking an example. So on the graph that you are seeing on your slide it shows the number of overall voters or vote shares secured by candidates in the 2016 US election.

You have Hillary Clinton securing 48% votes Donald Trump 46% and others some 6%. This is just for representation purpose. So now if you can see that Hillary Clinton has got more votes.

Now what you want to do is to understand how this voting pattern remains once we filter or drill down. Drilling down means you put one more dimension and see how the data is behaving what are the patterns there. We are first trying to understand this on two dimensions.

One is gender then by ethnicity. So there are two ways to achieve this. Either you can start with the drill down first you see the gender then within gender ethnicity or you can start with ethnicity and look down at the gender.

So let us see what happens when we do this two orders. So if you first drill down by gender you can see that 53% of the women voters have supported Hillary Clinton compared to 42% for the 4th Donald Trump. Now do it and then do the gender ethnicity drill down.

So 94% of the African women have voted for Hillary Clinton compared only to 4% for Donald Trump. So finally you will know that 94% of the women African voters have voted for Hillary Clinton and 4% is for Donald Trump. Now reverse the drill sequence first you go on ethnicity.

89% of the African Americans have voted for Hillary Clinton then you do the gender you would end up with the same result that you would get 94% of the African Americans women voting for Hillary Clinton. The end result is same but the path in which you have come to the end result matters. In the first case you are doing gender then the ethnicity the second case you are doing the ethnicity then gender.

You are reaching the same result but the path in which you drive the result may also it might create an emphasis on a particular narrative that you want to drive. So if I were to look at the first thing it would be like women voters preferred Hillary Clinton and African women have preferred it much more. Then if you just look at the second it is like African Americans have preferred Hillary and of them women also have preferred much more.

So both are two different narratives that you want to put out. So the order in which you do the drill down is also important and it can at times force particular narratives on to the audience and a good thing would be to enable the sequential drill down by both sides to the audience so that they are unbiased in their own conclusions. So finally what we have is the cherry picking or using decorative misuse.

Sometimes the analysts are over enthusiastic and want to present a particular view. They might take particular points or particular sections of the data visualisation and

highlight it and so as to show that particular aspect in a disproportionate way that catches the attention of the audience and drives the narrative in that direction. So this is not the best use of the visualisation space.

So you have to you should not cherry pick the data. Provide the data on the larger timeline or on the larger wider variety of the case so that you are not emphasising or deemphasizing a particular point. So let us talk this about a particular case and we will try to showcase the point.

So we have to prioritise accuracy over aesthetics. So look at the charts that you have on the slide. These charts indicate UK national debt as percentage of GDP.

So we are just in general talking about the debt as on January 26 being very high. So the chart on the left is a particular, it is cherry picked data, it is truncated so if you notice they are talking only from 1995 to 2016 where UK has existed much longer. UK economy was there from much longer.

They are just looking at the recent data and shown that the debt has been increasing. If one is looking at this graph, the immediate conclusion one would get is the data there is heavy increase in national debt as percentage of GDP and it is highest at the current level. But the moment you put the historical data like the ones before 1995, the debt levels are not very high as compared to the previous periods in which UK has experienced more than 200% debt.

So though there is a recent increase, nobody is denying that fact. By only selectively presenting data after 1995, the analyst is blowing out of proportion the actual numbers. But when seen in the entire historical context, this will create a sense that GDP has increased substantially.

So cherry picking of data. So one has to be very careful about the axis and how the data is presented by reading the information. And as an analyst, maybe you should give a caveat saying that this is only for this period or give some average numbers.

If you don't intend to present the entire earlier data, you might give a text box adding more context so that misleading information is not picked up by your audience. Then finally we see the misuse of colour saturation trying to mislead people with the use of darker colours in the visual. Let us say this has to do with the number of COVID positive cases across United States.

So if you see the second largest colour saturation legend, you would notice the darker shade is given to 5 lakh plus cases while the lighter shade is given to 1 lakh plus cases related to the lighter shade. So what is happening? So though you have Illinois, which is in Illinois, California and Texas which have 1 lakh plus cases, visually when you look at the graph, your attention would go to Wyoming, Minnesota and Pennsylvania and



Arizona than to Texas or Florida or California or New York where the cases are actually more. So this is counterintuitive use of colour saturation and should be avoided at the most.

The colour saturation should reflect the intensity that you want to present on the graph and that care has to be taken and at times one has to be careful of the legend while interpreting the graphs. Thank you.

-----End-----